

MAX Function in Excel

1. What is MAX?

The MAX function returns the largest number in a set of values. It solves the problem of finding the highest value in a long list, such as the highest sale, the top score, or the biggest salary. A Data Analyst uses it to find peak values and to compare other figures against the best result.

2. Syntax

`=MAX(number1, [number2], ...)`

Argument	Required?	What it accepts
number1	Required	A number, a cell reference, or a range (for example B2:B5)
number2, ...	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

Most often you will use a single range, such as `=MAX(B2:B5)`. You can also mix a range with a number: `=MAX(B2:B5, 60000)` returns whichever is larger, the highest value in the range or 60000.

3. Simple Example

Find the highest sales figure:

	A	B
1	Employee	Sales
2	Rahim	25000
3	Karim	42000
4	Hasan	35000
5	Nabil	50000
6	Maximum	50000

Formula in B6: `=MAX(B2:B5)`

Excel compares the four values in B2 to B5 (25000, 42000, 35000, 50000) and returns the largest one. **Result: 50000**

4. Another Example

Find the highest mark. One student was absent and one mark is missing:

	A	B
1	Student	Marks
2	Aisha	82
3	Rahim	Absent
4	Nabila	91
5	Tanvir	(blank)
6	Sadia	77
7	Highest	91

Formula in B7: `=MAX(B2:B6)`

Excel ignores the text "Absent" and the blank cell, and compares only 82, 91, and 77. **Result: 91**

5. When to Use It

- Find the highest sales figure or the best-selling product value
- Find the top exam score in a class
- Find the highest salary in a department
- Find the highest temperature, price, or expense in a period
- Find the most recent date in a list (dates are stored as numbers)

6. Important Things to Remember

- Text and empty cells inside a range are ignored. They are not counted as zero.
- A cell containing 0 is counted. Among negative numbers, the largest is the one closest to zero, so -5 is larger than -20.
- If a referenced cell contains an error such as #N/A, the whole MAX result shows that error.
- Numbers stored as text (often left-aligned) are ignored in a range. Convert them to real numbers first.
- For dates, MAX returns the latest date. Format the result cell as a date to read it.
- If the range has no numbers at all, MAX returns 0.

7. Common Mistakes

Mistake: Selecting an incomplete range, such as leaving out the last row. **Fix:** Check the highlighted range after typing the formula.

Mistake: Including a Total row in the range, so MAX returns the total. **Fix:** Select only the data rows.

Mistake: Getting a wrong maximum because some numbers are stored as text. **Fix:** Convert them to numbers (for example with Text to Columns) and check that they right-align.

Mistake: Expecting MAX to show who has the highest value. **Fix:** MAX returns only the number. Use a lookup to find the name (see the tip below).

8. Data Analyst Tip

MAX gives the value, not the name. To find who has it, combine it with a lookup: `=INDEX(A2:A5, MATCH(MAX(B2:B5), B2:B5, 0))` returns Nabil for the first example. It is also good practice to check your range with `=COUNT(B2:B5)` so you know how many numbers MAX actually compared.